

Static Bench — Field Setup Sheet

DWG paper4-rung1 · rev A

Paper 4 · Rung 1, the atomic test: does a wall echo appear in ESP32 CSI? Two boards, one metal plate, a straight corridor. Everything measured with a tape.

METHOD Monostatic-in-effect bistatic RCS w/ bkgnd sub.	BANDWIDTH 40 MHz (HT40)	RANGE CELL 3.75 m (c/2B)	GROUND TRUTH Tape measure
TX BOARD 28:84:85:48:40:20	RX BOARD 28:84:85:53:99:DC	BASELINE B 0.50 m	PLATE D (START) 12.0 m

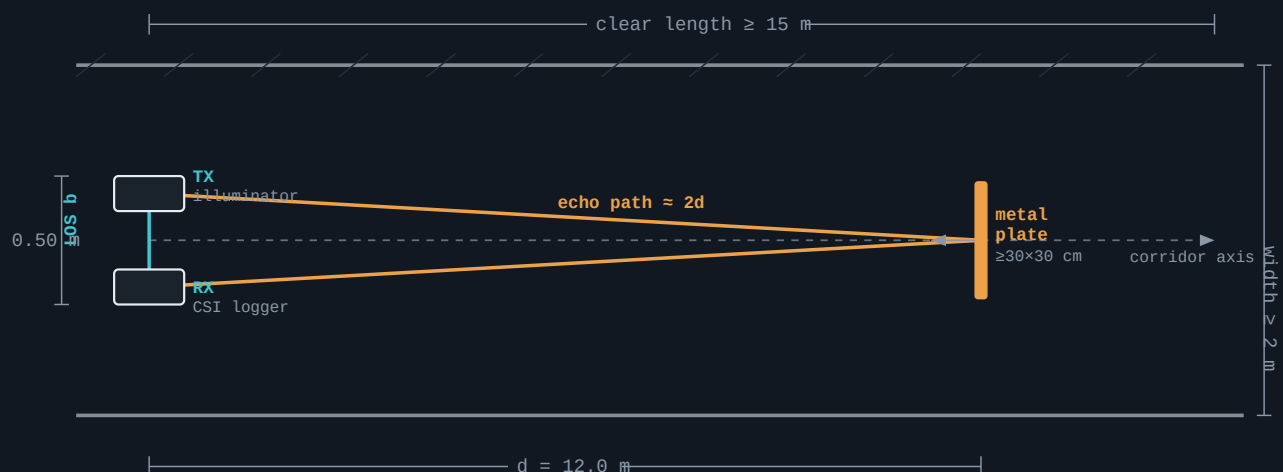
DRAWING 1 OF 4 — PLAN

A Plan view — looking down from above

The two boards sit together at the near end, **0.50 m** apart across the corridor. The plate stands on the corridor axis **12.0 m** away, facing the boards. Length is drawn to scale; the cross-corridor spacing is exaggerated so the two boards and their signal paths are legible.

Top-down geometry

scale: length 1:~ · width n.t.s.



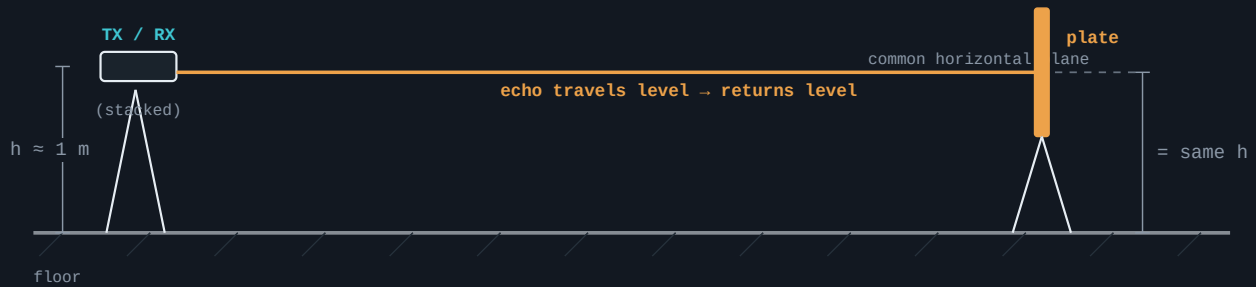
The 2.4° offset from the plate to each board (0.25 m over 12 m) is negligible – a flat plate square to the axis returns the echo to both boards.

B Side elevation — heights, co-planar

The critical height rule: **the boards' antennas and the centre of the plate sit at the same height**, so the reflection stays in one horizontal plane and the echo comes straight back. A tripod or a stack of boxes works — just measure it.

Vertical arrangement

heights n.t.s. · keep coplanar



Exact height doesn't matter; matching the two heights does. ~1 m off the floor keeps the LOS and echo clear of floor/ceiling clutter.

c Where the numbers come from

Path lengths

$b \ll d$



Echo path $\approx 2d$. Excess over LOS = $2d - b$.

The measured quantity

excess delay

$$c = 2.998 \times 10^8 \text{ m/s}$$

$$B = 40 \text{ MHz}$$

$$\text{Excess path} \quad 2d - b = 23.5 \text{ m}$$

$$\text{Excess delay} \quad (2d - b)/c = 78 \text{ ns}$$

$$\text{Path cell } (c/B) \quad 7.49 \text{ m}$$

$$\text{In cells} \quad 3.13$$

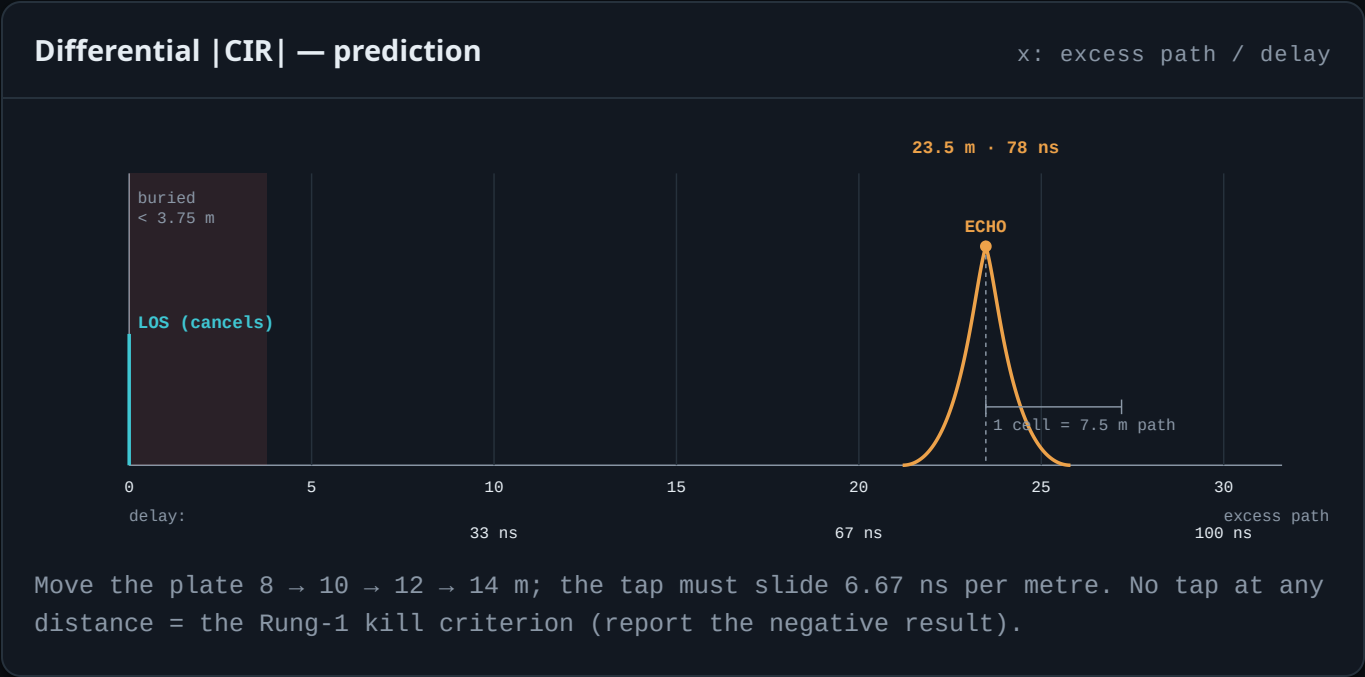
$$\text{Range cell } (c/2B) \quad 3.75 \text{ m}$$

This is the tap the differential should reveal.

DRAWING 4 OF 4 — EXPECTED RESULT

D What you should see (differential delay profile)

After **plate-in minus plate-out**, the direct path cancels and a single tap remains at the echo's excess delay. Anything closer than one range cell (**3.75 m**) is swallowed by the direct path — which is why the plate starts at 12 m.



BILL OF MATERIALS

E Elements & dimensions

ITEM	SPEC / DIMENSION	QTY	NOTES
ESP32-S3 board	DevKitC-1, HT40 CSI, on COM/UART port	2	TX + RX; ~\$16
Metal plate	flat, conductive, ≥ 30 × 30 cm	1	baking tray; flat = specular
Tape measure	≥ 15 m (laser ± cm better)	1	the ground-truth instrument
Tripods / stands	hold boards + plate at one height	2	coplanar is the rule
USB cables + laptop	data-capable; venv installed	1	runs the capture + analysis

ITEM	SPEC / DIMENSION	QTY	NOTES
Corridor (site)	straight ≥ 15 m, width ≥ 2 m	1	the make-or-break requirement

baseline b = 0.50 m

plate d = 8 / 10 / 12 / 14 m

height h = same for all

channel = 2.4 GHz #1

records = 1000 / recording

FINDING A LOCATION

F Site requirements — what to look for

- ☐ **Straight, clear ≥ 15 m.** A long hallway, warehouse aisle, empty parking lane, or gym.
- ☐ **Nothing within ~ 4 m** of the boards along the axis (closer than one range cell hides in the direct path).
- ☐ **Room to move the plate** to 8, 10, 12, 14 m and mark each spot.
- ☐ **Quiet during capture** — no people/carts crossing the path while recording (~ 10 s each).
- ☐ **Power** — a socket or a laptop battery; the boards run off USB.

Why not a small room? At 40 MHz the range cell is 3.75 m. In a 4–5 m room every wall echo lands in the same bin as the direct path and the differential shows nothing — you'd measure *nothing*, not a null result. The site size is physics, not preference.

ON-SITE PROCEDURE

G Setup & run — hit-run steps

- 01 Place the boards** at the near end, 0.50 m apart, both facing down the corridor. Plug both into the laptop via their **COM/UART** ports. They power on and the TX starts transmitting automatically.
- 02 Confirm the link (Rung 0.5).** Boards can be on a table for this. Run the check — it auto-finds the port and confirms one clean tap:

```
# from the repo root
.venv/bin/python experiments/run_bench.py check
```

- 03 Set the geometry.** Boards 0.50 m apart; plate on the axis at 12 m, at the same height, facing the boards. Measure each with the tape.

04

Run the atomic test. It prompts you to record with the plate in, then to remove it and record out — twice — then prints the detected echo vs the prediction:

```
.venv/bin/python experiments/run_bench.py rung1 --plate 12
```

05

Read the result. A tap near **23.5 m / 78 ns** = success (first delay-resolved reflector from an ESP32). Then repeat at 10 and 14 m to confirm it tracks.

06

If nothing appears at 10, 12 *and* 14 m — that is the honest Rung-1 kill criterion. Save the captures; the negative result is itself a finding.

Captures auto-save to data/hw_captures/ with timestamps, so every run is reproducible offline with experiments/run_hw_phantom.py.

wifi-radar · paper 4 · branch paper4-hardware-testbed

Rung 1 setup · DWG rev A · tape-measured ground truth